

FITBUDDY

AI Fitness Plan Generator using Gemini Models

PROJECT DEMONSTRATION PHASE

1. Demonstration Overview

The demonstration phase shows how FitBuddy works from user input to AI-generated fitness plan. The purpose of the demonstration is to explain the application's workflow, major features, Gemini integration, and the final output in a simple and practical manner.

2. Demonstration Objective

The main objective is to demonstrate that FitBuddy can collect user requirements and use Gemini models to generate a structured fitness plan. The demo should show the complete flow rather than only individual screens.

3. Key Features to Demonstrate

- Simple user input interface
- Fitness goal selection
- Experience-level selection
- Workout frequency and available time
- Workout preference and equipment information
- AI-based plan generation using Gemini
- Clear day-wise fitness plan output
- Error handling for incomplete inputs

4. Demonstration Setup

Component	Purpose
FitBuddy Application	Main application used for the demonstration.
User Input Form	Collects fitness requirements.
Gemini Model/API	Generates the personalized plan.
Internet/API Connection	Allows communication with the AI service when applicable.

5. Demo Flow

Open Application → Enter Fitness Details → Validate Inputs → Generate Plan → Gemini Processes Request → Receive AI Response → Display Personalized Fitness Plan

PAGE 2 – USER INPUT DEMONSTRATION

6. Step 1 – Opening FitBuddy

The demonstration begins by opening the FitBuddy application. The home screen introduces the purpose of the application and provides an option to start generating a fitness plan.

7. Step 2 – Entering Fitness Goal

The user selects or enters the main fitness goal. This information helps the AI understand the type of plan that needs to be generated.

8. Step 3 – Selecting Experience Level

The user provides an experience level such as beginner, intermediate, or advanced. This input can be used to adjust the complexity of the generated routine.

9. Step 4 – Providing Workout Availability

The user enters the number of available workout days and approximate time available for each session. FitBuddy uses this information to create a practical schedule.

10. Step 5 – Preferences and Equipment

The user can provide workout preferences and information about available equipment. This helps the generated plan better match the user's situation.

Sample Demonstration Input

Field	Sample Value
Fitness Goal	General fitness / strength improvement
Experience Level	Beginner
Workout Days	4 days per week
Session Time	30 minutes
Workout Preference	Home workout
Equipment	Minimal / available home equipment

11. Input Validation

- Check that required fields are completed.
- Check that values are in an acceptable format.
- Show a clear message when required information is missing.
- Allow the user to correct the input before generation.

PAGE 3 – GEMINI AI DEMONSTRATION

12. AI Generation Step

After the user submits the information, FitBuddy prepares the details as an AI request. The request is sent to the Gemini model through the application's configured integration. The model processes the information and generates a fitness plan.

13. Prompt Structure Demonstration

A suitable request can contain the user's goal, experience level, available days, session duration, preferences, and equipment. The prompt can also instruct the model to return the answer in a structured day-wise format.

14. Gemini Processing Flow

Stage	Action
Input	User fitness details are collected.
Prompt	Application prepares a structured request.
AI Processing	Gemini interprets the request and generates content.
Response	Generated fitness plan is returned.
Display	Application formats and shows the plan.

15. Example Generated Output Structure

- Day 1 – Workout focus and exercise list
- Day 2 – Workout focus and exercise list
- Day 3 – Rest/recovery or suitable activity
- Day 4 – Workout focus and exercise list
- Basic session structure such as warm-up, main workout, and cool-down
- Simple instructions for suggested exercises

16. Regeneration Demonstration

If the application supports regeneration, the user can change an input or request another suitable plan. The updated request is sent to Gemini and a new response is displayed. This demonstrates the flexibility of generative AI within FitBuddy.

PAGE 4 – OUTPUT & FEATURE DEMONSTRATION

17. Displaying the Fitness Plan

The generated response should be displayed in an organized format. Headings, day-wise sections, exercise names, and simple instructions should be separated clearly so that the user can understand the result.

18. Main Output Elements

- Weekly workout schedule
- Exercise names and basic guidance
- Approximate workout duration
- Rest or recovery days
- Plan matched to the submitted preferences
- Simple and readable formatting

19. Error Scenario Demonstration

Scenario	Expected Application Behavior
Required field is empty	Ask the user to complete the field.
Invalid input	Show a correction message.
Gemini/API unavailable	Display a friendly error and avoid application crash.
Unexpected response	Handle the response safely and provide a retry option if available.

20. Feature-by-Feature Demo Checklist

- Launch application
- Show home/interface screen
- Enter user details
- Submit valid inputs
- Generate AI fitness plan
- Display generated result
- Test incomplete input
- Demonstrate error handling
- Regenerate or modify plan if supported

21. User Experience During Demonstration

The demonstration should emphasize that the application is simple to use. The presenter should explain what information is entered, why the information matters, how Gemini is used, and how the final result is displayed.

PAGE 5 – DEMO RESULTS, TESTING & CONCLUSION

22. Demonstration Result

The expected demonstration result is a successfully generated fitness plan based on the information entered by the user. The output should be structured, readable, and clearly connected to the selected requirements.

23. Demonstration Testing

Test	Expected Result
Valid input	Fitness plan is generated.
Different goal	Generated plan reflects the changed goal.
Different experience level	Plan difficulty can change according to the input.
Different available time	Plan structure reflects the selected duration.
Missing input	User receives a validation message.
AI/API error	Application handles the error without crashing.

24. Key Points to Explain During Presentation

- What problem FitBuddy solves.
- Why personalized planning is useful.
- What information the user provides.
- How Gemini is integrated into the application.
- How the prompt is prepared.
- How the AI-generated response is displayed.
- What limitations the AI-generated plan has.
- What features can be added in future.

25. Safety and Responsible Demonstration

FitBuddy should be demonstrated as an AI-assisted planning tool. The generated plan should not be presented as medical advice or as a guaranteed solution for health or fitness outcomes. Users with injuries, medical conditions, or specific health concerns should seek appropriate professional guidance.

26. Conclusion

The Project Demonstration Phase proves the practical workflow of FitBuddy: collecting user information, preparing an AI request, generating a response through Gemini, and presenting the result in a clear fitness-plan format. The demonstration provides a complete view of how the proposed system can work in a real application.

Project Name: FitBuddy – AI Fitness Plan Generator using Gemini Models

Phase: Project Demonstration Phase

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